

# Blinkit Customer Segmentation & Business Strategy Report

End-to-End RFM Feature Engineering & K-Means Clustering Machine Learning Project

## 1. Executive Summary & Segmentation Architecture

This project applies unsupervised machine learning to Blinkit's quick-commerce order data. By transforming transaction records into Recency, Frequency, and Monetary (RFM) behavioral profiles, a K-Means model (k=4) partitions users into actionable cohorts:

### Champions (VIP)

**Profile:** Highest purchase frequency & spend. Drives >50% of metro revenues.  
**Strategy:** Exclusive loyalty perks, early access to flash sales, priority delivery.

### Loyal Customers

**Profile:** Consistent repeat buyers with steady order volumes.  
**Strategy:** Subscription bundles (Blinkit Pass), pantry staple cross-selling.

### At-Risk / Lost

**Profile:** Long inactivity period; high churn risk. ~Rs 2.4L trapped in Delhi.  
**Strategy:** Win-back discount triggers, personalized push alerts, cart reminders.

### New / Recent

**Profile:** Recent first-time buyers with low initial frequency.  
**Strategy:** 2nd and 3rd order discount vouchers to accelerate habit formation.

## 2. Key Data Insights & Strategic Action Items

| Business Metric            | Observed Finding                                                                   | Recommended Business Action                                                                       |
|----------------------------|------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| City Revenue Concentration | Delhi (Rs 17.69L VIP) & Bangalore (Rs 14.59L VIP) generate maximum revenue.        | Allocate top ad spends & inventory safety stock to Delhi & Bangalore micro-warehouses.            |
| Payment Gateway Dominance  | UPI drives >70% volume across all segments; COD follows at ~17%.                   | Optimize UPI 1-click checkout flow & partner with UPI apps for cashback campaigns.                |
| Delivery Latency vs Churn  | Average delivery time is uniform (~29.4 to 29.6 mins) across all customer cohorts. | Delivery delay is not the root cause of churn; focus retention on pricing and stock availability. |

### Resume / Portfolio Highlight:

"Engineered an end-to-end Customer Segmentation model using RFM feature extraction and K-Means Clustering on quick-commerce transaction records. Identified core growth hubs (Delhi & Bangalore driving 55%+ VIP sales) and proved churn was decoupled from delivery latency (~29.5m avg), providing data-backed retention roadmaps."